

W4D3 — 2-Page Summary: Feature Engineering, CV & Model Comparison

1. Engineered Feature List

Feature	Type	Creation Rule	Justification	MI Score
age_bucket	categorical	pd.cut(age)	Age groups capture non-linear age-income effects	0.0599
hours_bin	categorical	pd.cut(hours-per-week)	Work-hour patterns relate to income	0.0379
cap_gain_flag	binary	capital-gain > 0	Any capital gains often indicate additional income sources	0.0296
log_cap_gain	numeric	log1p(capital-gain)	Reduce skew and limit outlier influence of large gains	0.0830
higher_education	binary	education in {Bachelors,Masters,Doctorate}	Higher education is correlated with higher earnings	0.0371

edu_hours_interaction	numeric	education-num * hours-per-week	Captures combined effect of education and hours worked	0.0815
net_capital	numeric	capital-gain - capital-loss	Net capital may be more informative than separate values	0.1212

All 7 features are built from current-row data only (no target encoding, no leakage).

2. Cross-Validation Comparison

Results Table (5-fold Stratified CV)

Model	Precision	F1	ROC AUC	Time (s)
HistGradientBoosting	0.7808 ± 0.0080	0.7122 ± 0.0070	0.9282 ± 0.0017	25.6
LogisticRegression	0.7498 ± 0.0050	0.6739 ± 0.0043	0.9129 ± 0.0023	19.2
RandomForest	0.7288 ± 0.0083	0.6735 ± 0.0079	0.9047 ± 0.0026	78.0

Key Observations

- **Best precision:** HistGradientBoosting (0.7808)
- **Best ROC AUC:** HistGradientBoosting (0.9282)
- See `cv_boxplots_all_metrics.png` for fold-level distributions.

3. Statistical Test Results

Top 2 models by precision: HistGradientBoosting vs LogisticRegression

Paired t-test $p=2.2760e-04$; mean gap= $+0.0309$. Statistically significant. Practically meaningful (>1 pp).

Wilcoxon $p=0.0625$ (confirms/does not confirm fold-level consistency depending on value above).

4. Feature Importances & Coefficients

Top engineered features (RandomForest importances):

feature	importance
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edu_hours_interaction	0.067944
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net_capital	0.052929
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log_cap_gain	0.041891
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age_bucket_total	0.038555
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higher_education	0.017558
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hours_bin_total	0.015508
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cap_gain_flag	0.009523
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- `log_cap_gain` and `cap_gain_flag` rank highest among engineered features — capital activity is the strongest non-demographic predictor of income.
- `edu_hours_interaction` confirms the multiplicative effect of education $\times$ hours worked.
- `higher_education` boolean is compact but highly informative.

5. Feature Selection

- SelectKBest (mutual information, $k=40$) used inside CV pipeline (no leakage).
- Performance vs full feature set: see printed output in Task 5 cell.
- **Decision:** Retain all 7 engineered features. Drop low-MI one-hot columns (sparse native-country/occupation sub-categories).

6. Recommended Models & Features for Day 4 Tuning

Model: `HistGradientBoosting` (highest precision; recommend also benchmarking `HistGradientBoostingClassifier`)

Feature set: All 7 engineered features + original numeric/categorical features, with SelectKBest k=40 as a pipeline step.

Suggested tuning parameters:

- RandomForest: `n_estimators`, `max_depth`, `min_samples_leaf`, `max_features`
- HistGradientBoosting: `learning_rate`, `max_iter`, `max_leaf_nodes`, `l2_regularization`